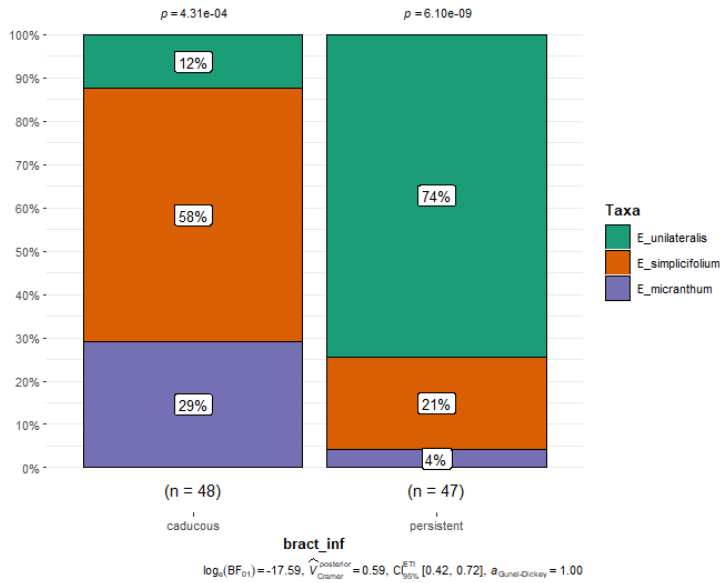


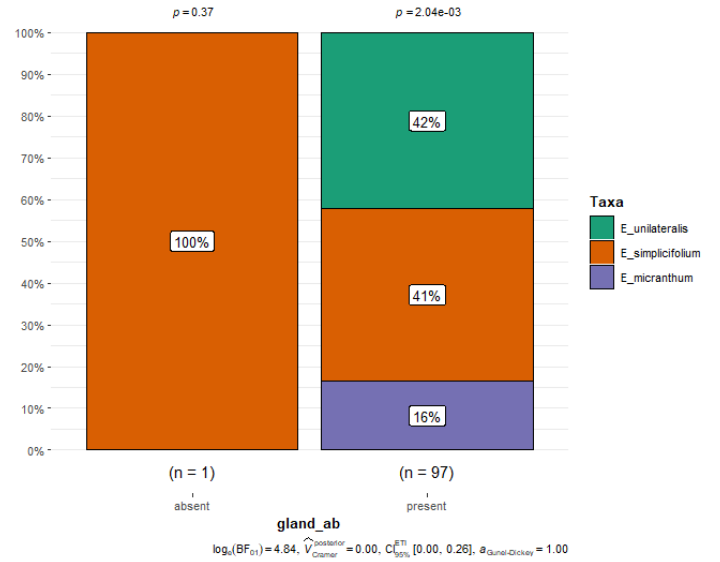
Association between bract_inf and Taxa

$\chi^2_{\text{Pearson}}(2) = 38.03$, $p = 5.51\text{e-}09$, $\hat{V}_{\text{Cramer}} = 0.62$, $\text{CI}_{95\%} [0.40, 0.82]$, $n_{\text{obs}} = 95$



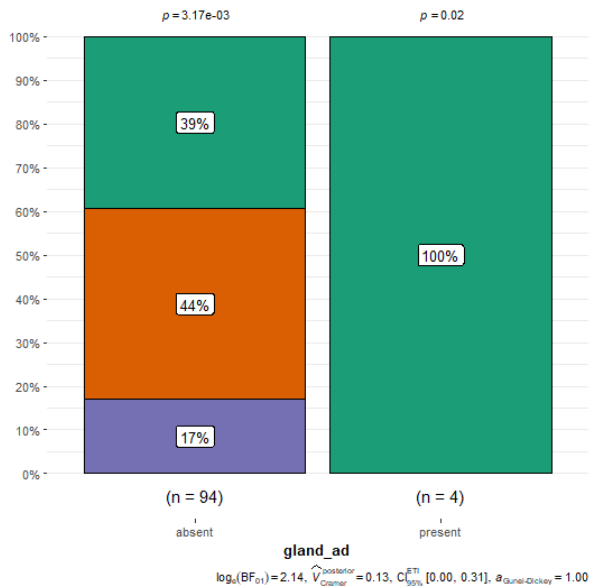
Association between gland_ab and Taxa

$\chi^2_{\text{Pearson}}(2) = 1.40$, $p = 0.50$, $\hat{V}_{\text{Cramer}} = 0.00$, $\text{CI}_{95\%} [0.00, 0.25]$, $n_{\text{obs}} = 98$



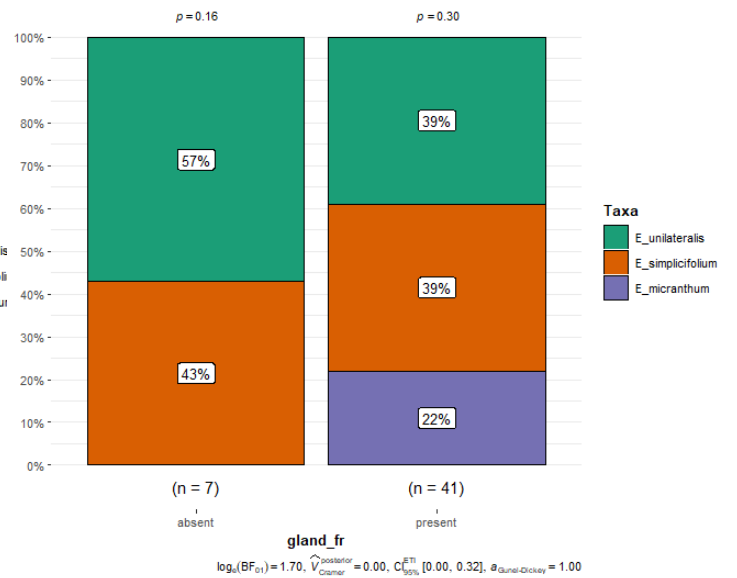
Association between gland_ad and Taxa

$\chi^2_{\text{Pearson}}(2) = 5.80$, $p = 0.06$, $\hat{V}_{\text{Cramer}} = 0.20$, $\text{CI}_{95\%} [0.00, 0.40]$, $n_{\text{obs}} = 98$



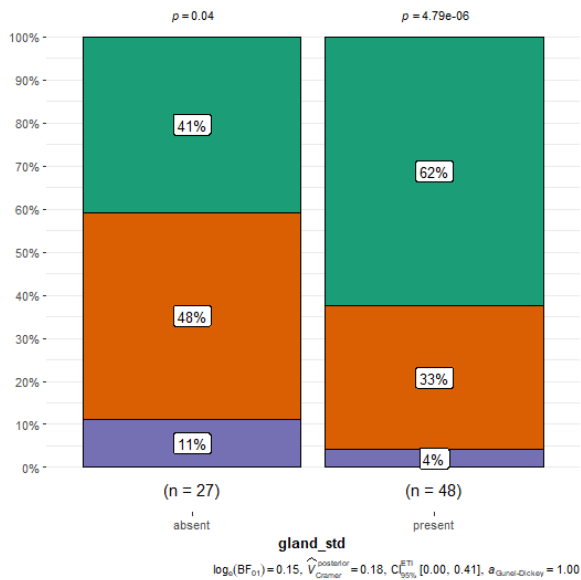
Association between gland_fr and Taxa

$\chi^2_{\text{Pearson}}(2) = 2.03$, $p = 0.36$, $\hat{V}_{\text{Cramer}} = 0.00$, $\text{CI}_{95\%} [0.00, 0.41]$, $n_{\text{obs}} = 48$



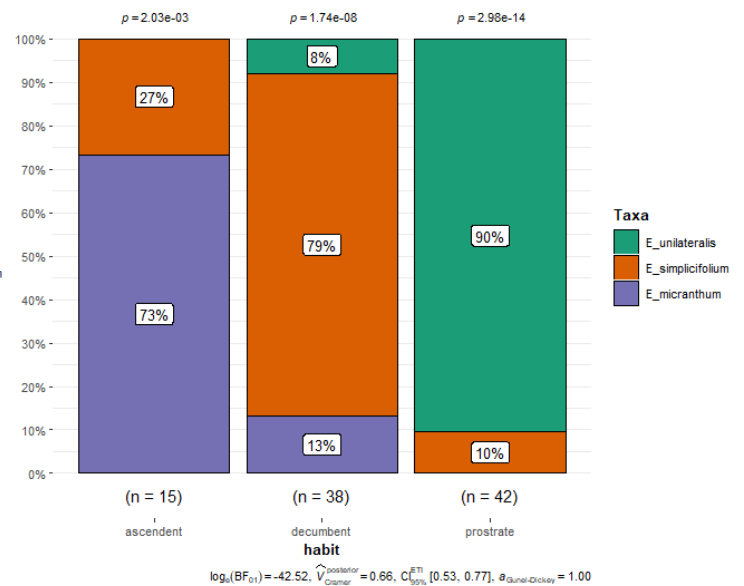
Association between gland_std and Taxa

$\chi^2_{\text{Pearson}}(2) = 3.73$, $p = 0.16$, $\hat{V}_{\text{Cramer}} = 0.15$, $\text{CI}_{95\%} [0.00, 0.40]$, $n_{\text{obs}} = 75$



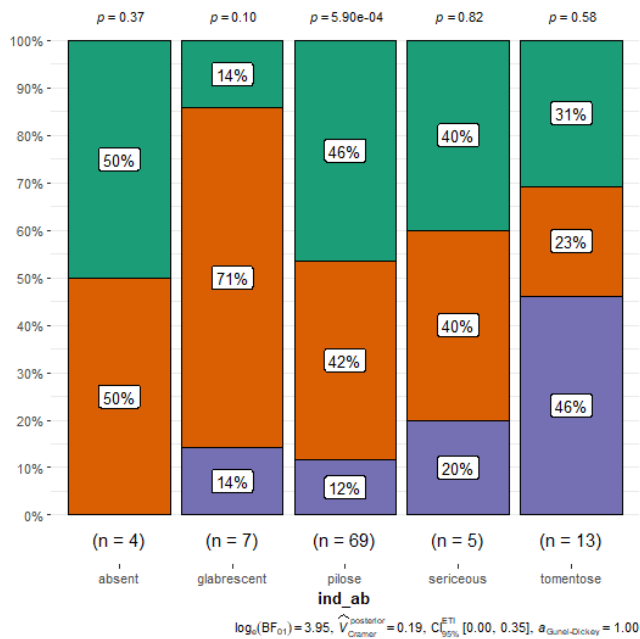
Association between habit and Taxa

$\chi^2_{\text{Pearson}}(4) = 99.84$, $p = 1.06\text{e-}20$, $\hat{V}_{\text{Cramer}} = 0.72$, $\text{CI}_{95\%} [0.56, 0.85]$, $n_{\text{obs}} = 95$



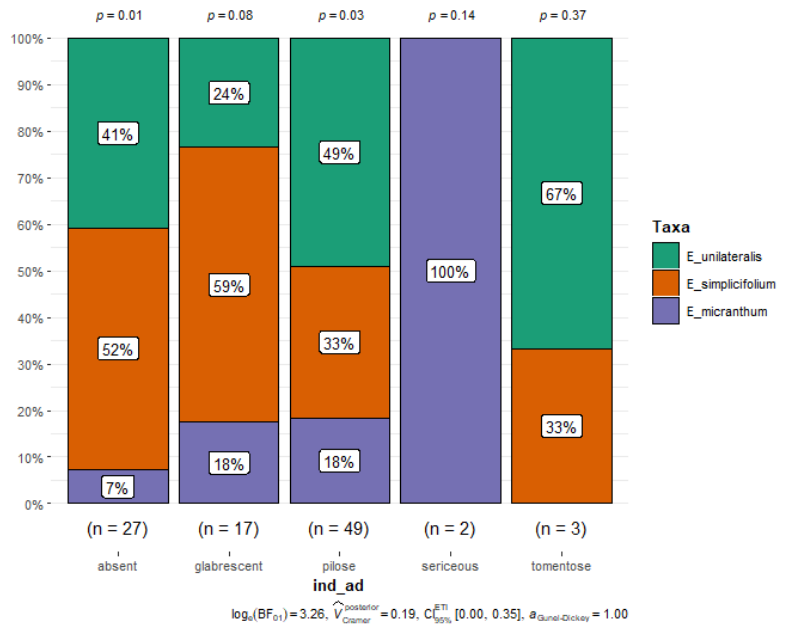
Association between ind_ab and Taxa

$\chi^2_{\text{Pearson}}(8) = 13.43, p = 0.10, \hat{V}_{\text{Cramer}} = 0.17, \text{CI}_{95\%} [0.00, 0.28], n_{\text{obs}} = 98$



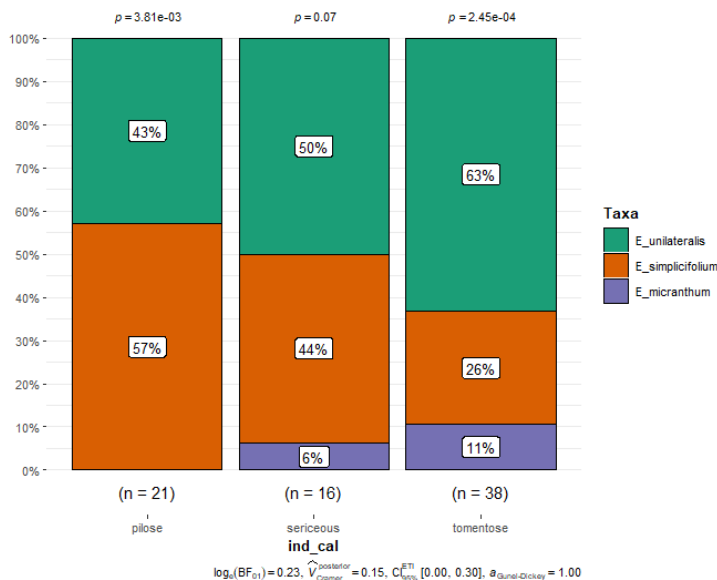
Association between ind_ad and Taxa

$\chi^2_{\text{Pearson}}(8) = 17.47, p = 0.03, \hat{V}_{\text{Cramer}} = 0.22, \text{CI}_{95\%} [0.00, 0.33], n_{\text{obs}} = 98$



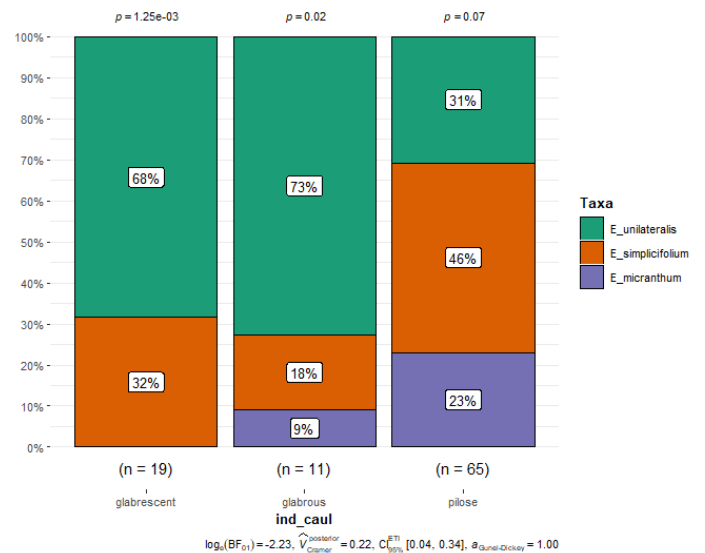
Association between ind_cal and Taxa

$\chi^2_{\text{Pearson}}(4) = 6.81, p = 0.15, \hat{V}_{\text{Cramer}} = 0.14, \text{CI}_{95\%} [0.00, 0.30], n_{\text{obs}} = 75$



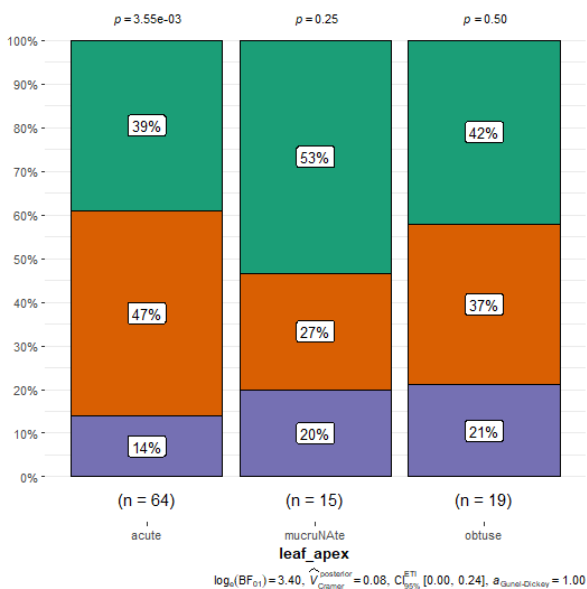
Association between ind_caul and Taxa

$\chi^2_{\text{Pearson}}(4) = 14.70, p = 5.36\text{e-}03, \hat{V}_{\text{Cramer}} = 0.24, \text{CI}_{95\%} [0.00, 0.37], n_{\text{obs}} = 95$



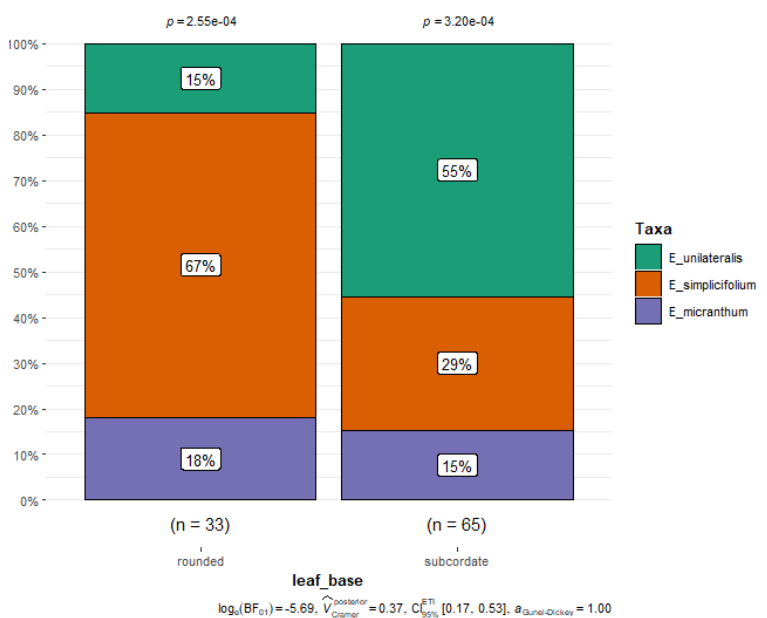
Association between leaf_apex and Taxa

$\chi^2_{\text{Pearson}}(4) = 2.50, p = 0.64, \hat{V}_{\text{Cramer}} = 0.00, \text{CI}_{95\%} [0.00, 0.14], n_{\text{obs}} = 98$



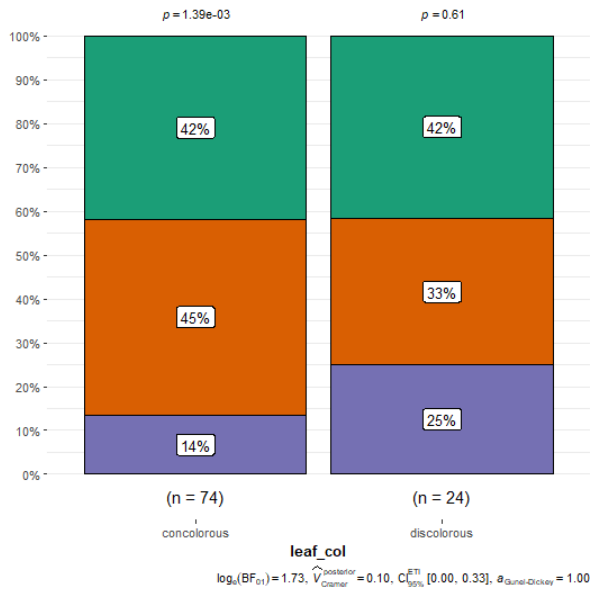
Association between leaf_base and Taxa

$\chi^2_{\text{Pearson}}(2) = 15.91, p = 3.52\text{e-}04, \hat{V}_{\text{Cramer}} = 0.38, \text{CI}_{95\%} [0.12, 0.58], n_{\text{obs}} = 98$



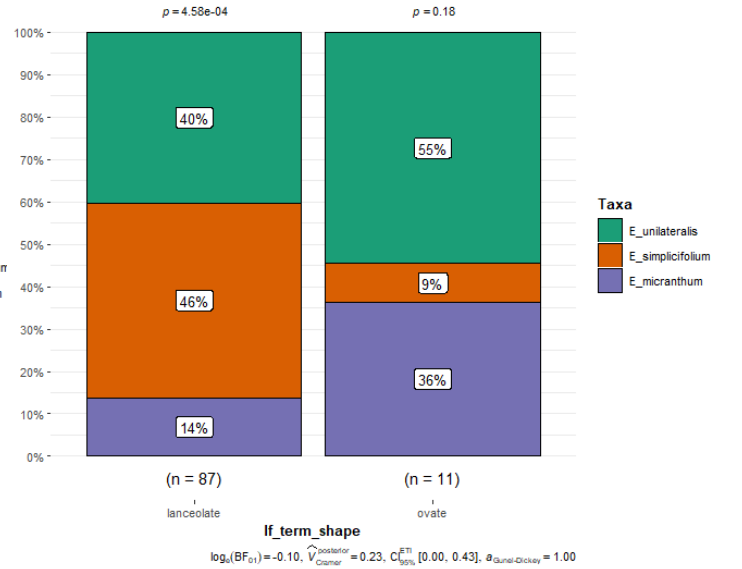
Association between leaf_col and Taxa

$\chi^2_{\text{Pearson}}(2) = 2.01, p = 0.37, \hat{V}_{\text{Cramer}} = 0.00, \text{CI}_{95\%} [0.00, 0.28], n_{\text{obs}} = 98$



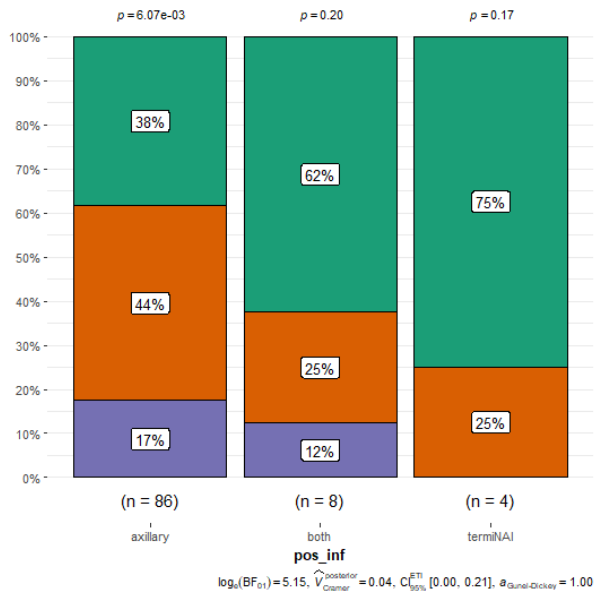
Association between lf_term_shape and Taxa

$\chi^2_{\text{Pearson}}(2) = 6.70, p = 0.04, \hat{V}_{\text{Cramer}} = 0.22, \text{CI}_{95\%} [0.00, 0.42], n_{\text{obs}} = 98$



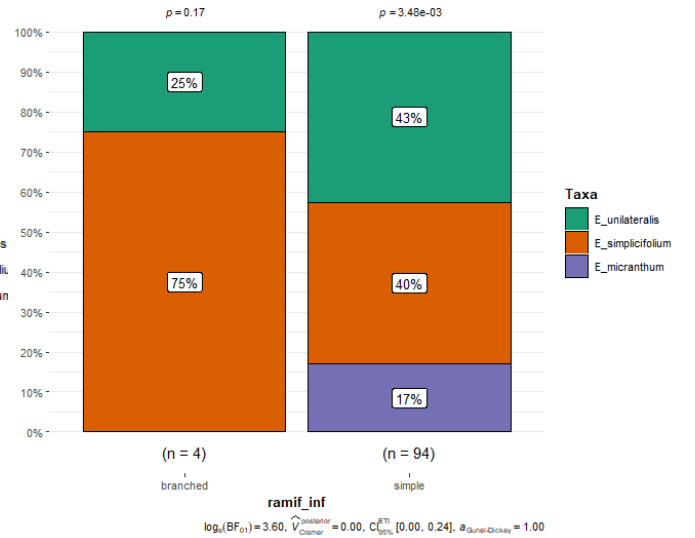
Association between pos_inf and Taxa

$\chi^2_{\text{Pearson}}(4) = 3.83, p = 0.43, \hat{V}_{\text{Cramer}} = 0.00, \text{CI}_{95\%} [0.00, 0.19], n_{\text{obs}} = 98$



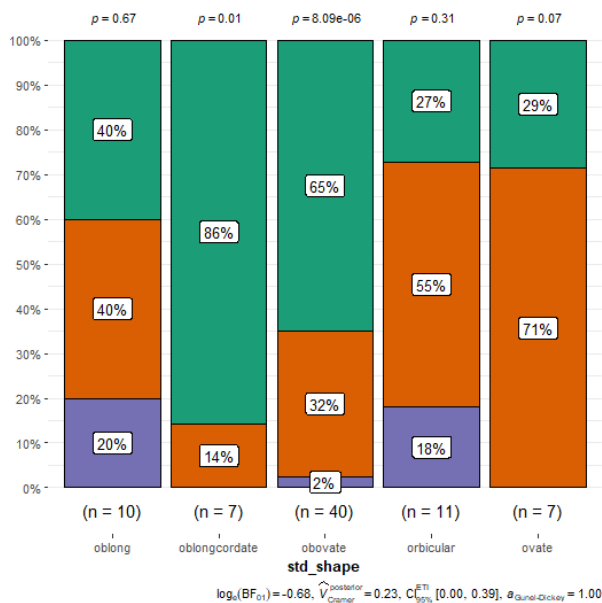
Association between ramif_inf and Taxa

$\chi^2_{\text{Pearson}}(2) = 2.06, p = 0.36, \hat{V}_{\text{Cramer}} = 0.02, \text{CI}_{95\%} [0.00, 0.29], n_{\text{obs}} = 98$



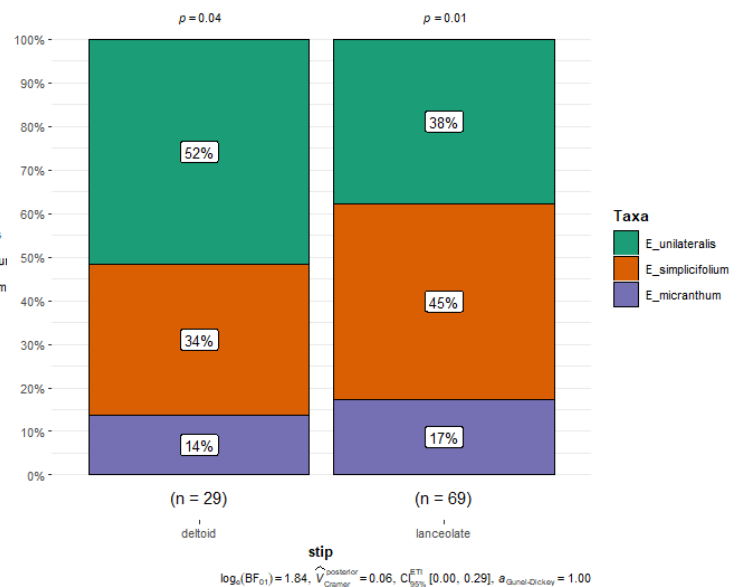
Association between std_shape and Taxa

$\chi^2_{\text{Pearson}}(8) = 15.76, p = 0.05, \hat{V}_{\text{Cramer}} = 0.23, \text{CI}_{95\%} [0.00, 0.36], n_{\text{obs}} = 75$



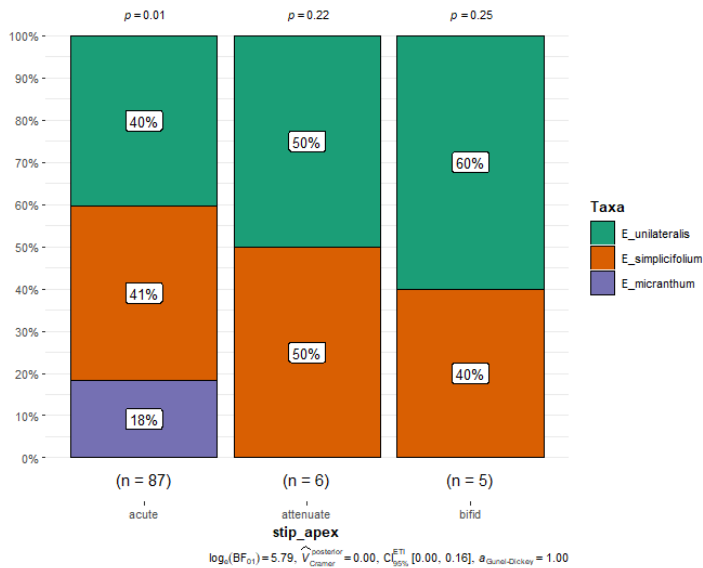
Association between stip and Taxa

$\chi^2_{\text{Pearson}}(2) = 1.66, p = 0.44, \hat{V}_{\text{Cramer}} = 0.00, \text{CI}_{95\%} [0.00, 0.27], n_{\text{obs}} = 98$



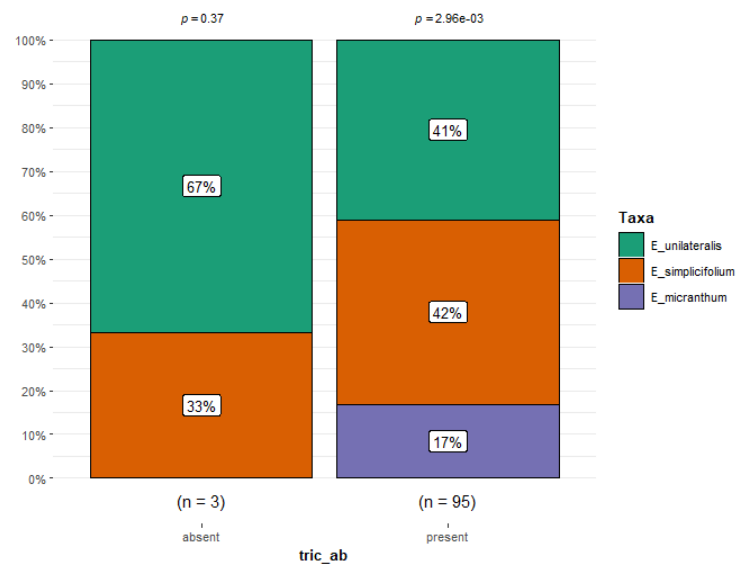
Association between stip_apex and Taxa

$\chi^2_{\text{Pearson}}(4) = 2.67, p = 0.61, \hat{V}_{\text{Cramer}} = 0.00, \text{CI}_{95\%} [0.00, 0.15], n_{\text{obs}} = 98$



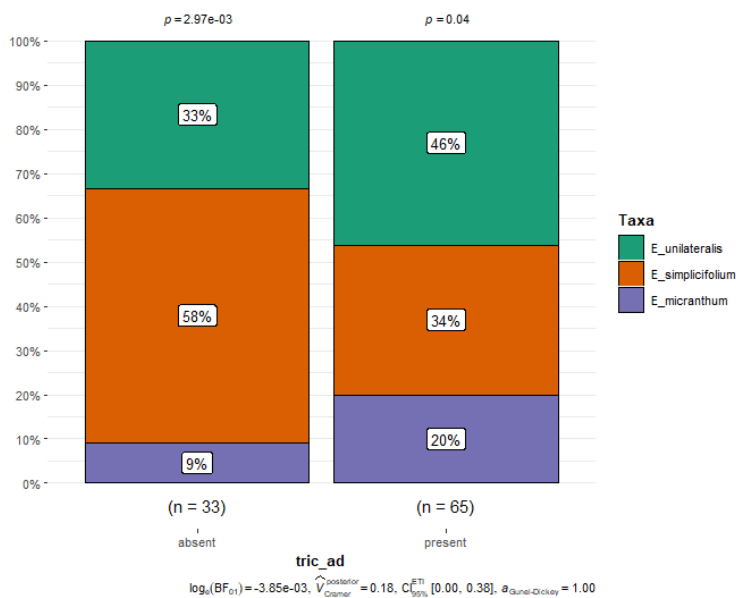
Association between tric_ab and Taxa

$\chi^2_{\text{Pearson}}(2) = 1.01, p = 0.60, \hat{V}_{\text{Cramer}} = 0.00, \text{CI}_{95\%} [0.00, 0.23], n_{\text{obs}} = 98$



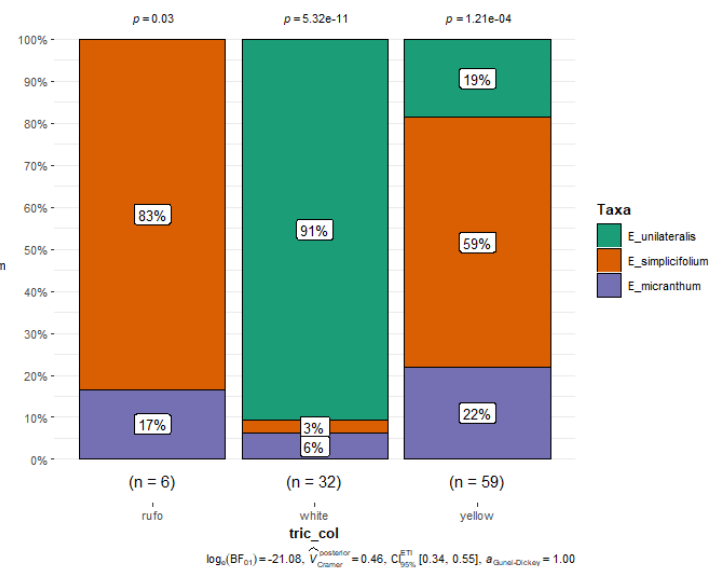
Association between tric_ad and Taxa

$\chi^2_{\text{Pearson}}(2) = 5.40, p = 0.07, \hat{V}_{\text{Cramer}} = 0.19, \text{CI}_{95\%} [0.00, 0.39], n_{\text{obs}} = 98$

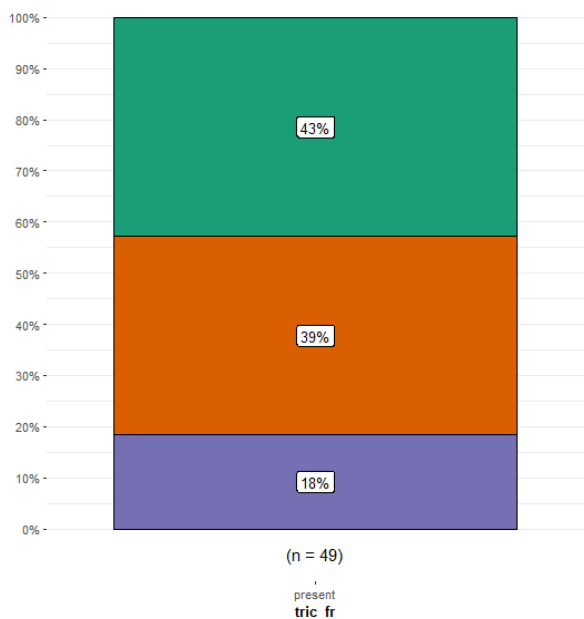


Association between tric_col and Taxa

$\chi^2_{\text{Pearson}}(4) = 49.89, p = 3.80\text{e-}10, \hat{V}_{\text{Cramer}} = 0.49, \text{CI}_{95\%} [0.32, 0.62], n_{\text{obs}} = 97$

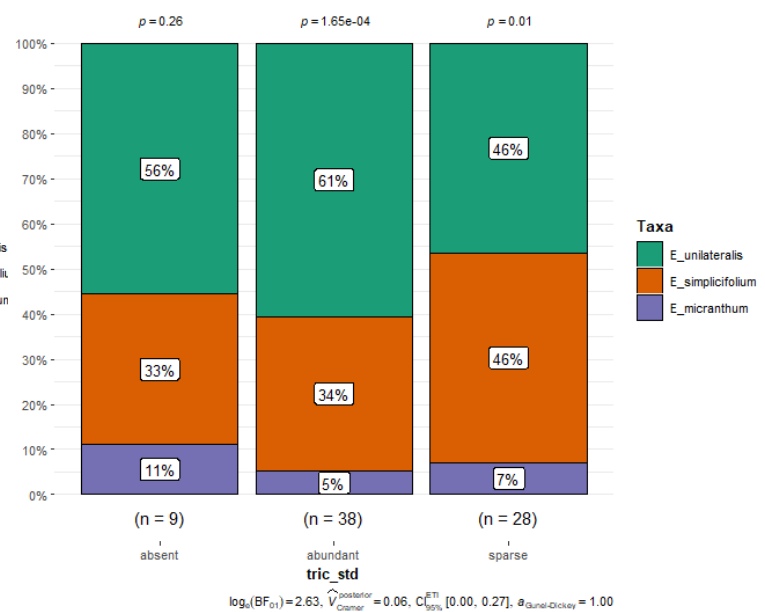


Association between tric_fr and Taxa

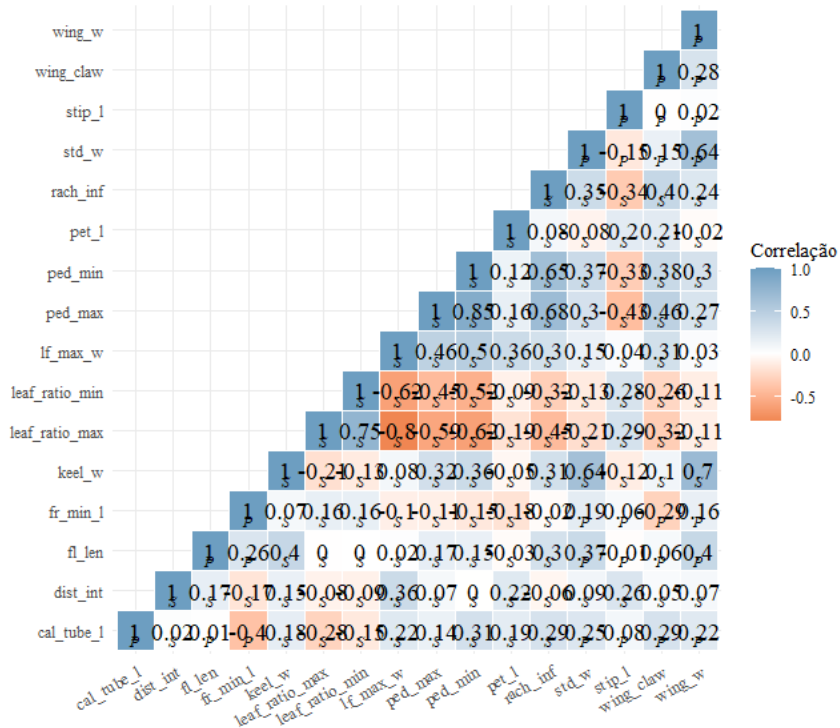


Association between tric_std and Taxa

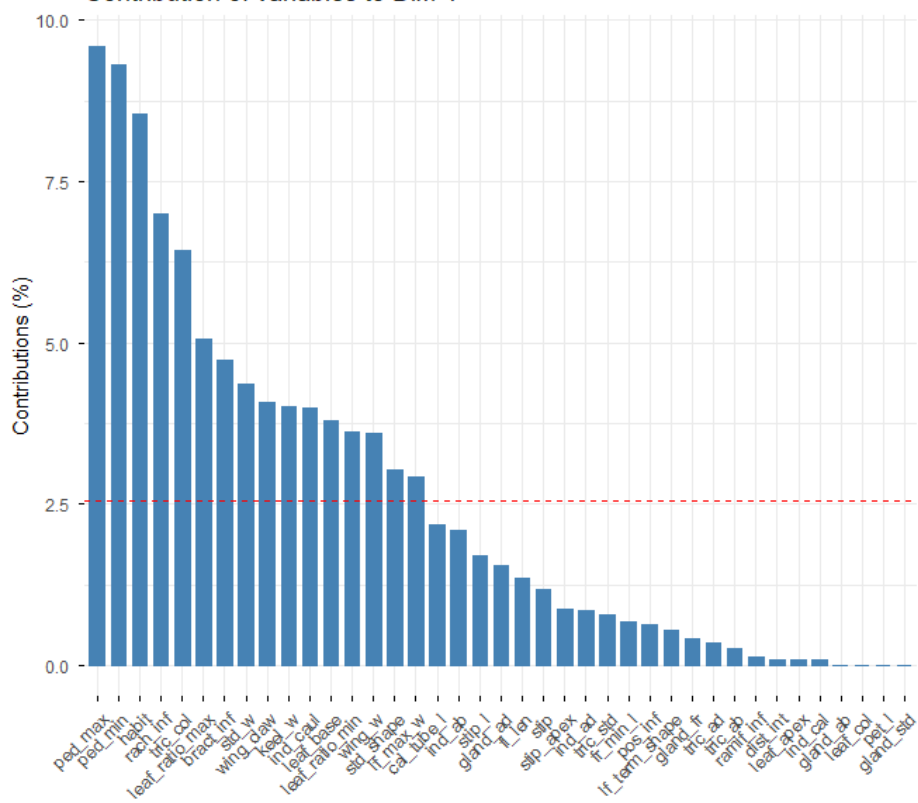
$\chi^2_{\text{Pearson}}(4) = 1.67, p = 0.80, \hat{V}_{\text{Cramer}} = 0.00, \text{CI}_{95\%} [0.00, 0.10], n_{\text{obs}} = 75$



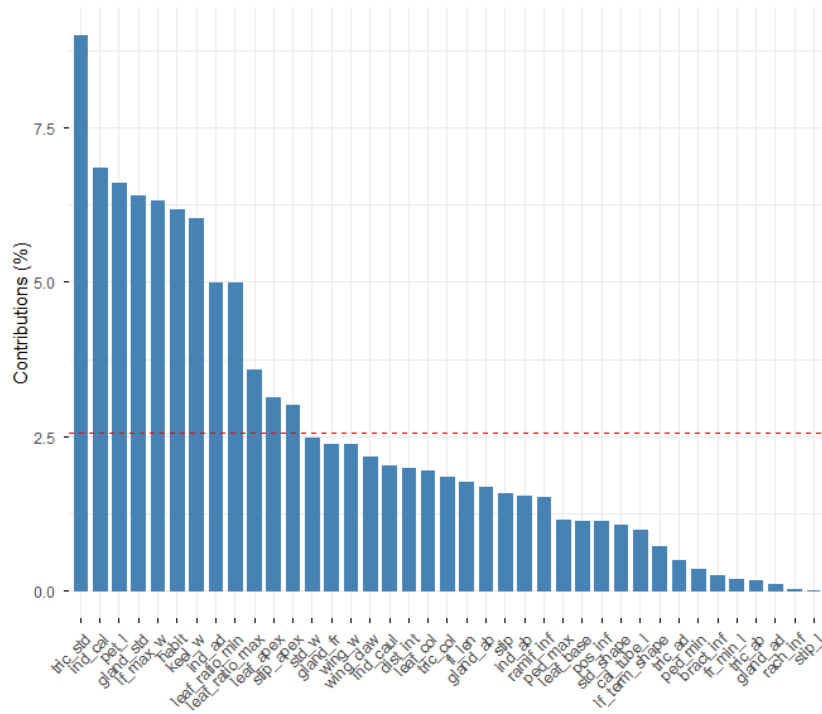
Matriz de Correlação com Tipo (P = Pearson, S = Spearman)



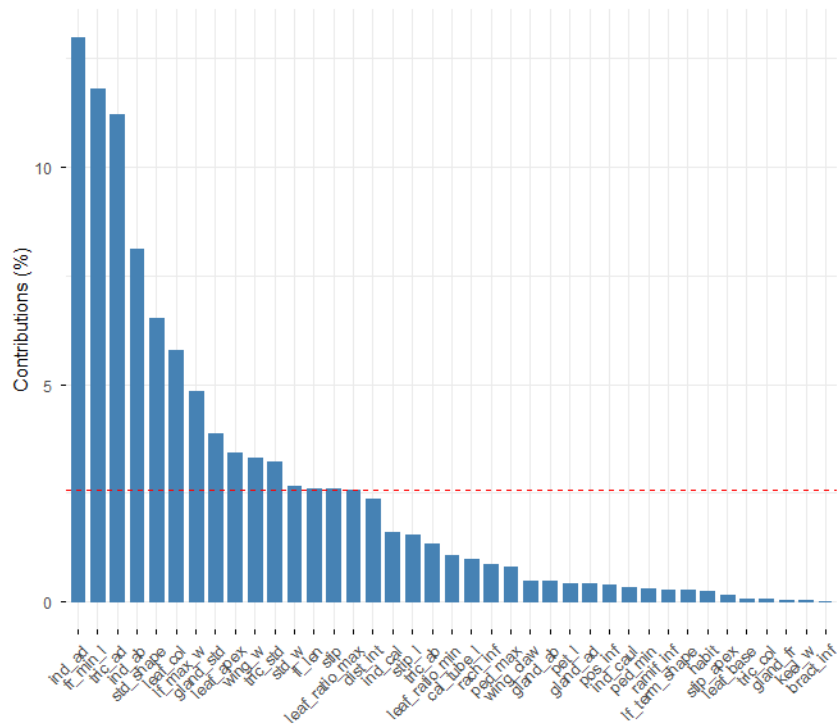
Contribution of variables to Dim-1



Contribution of variables to Dim-2

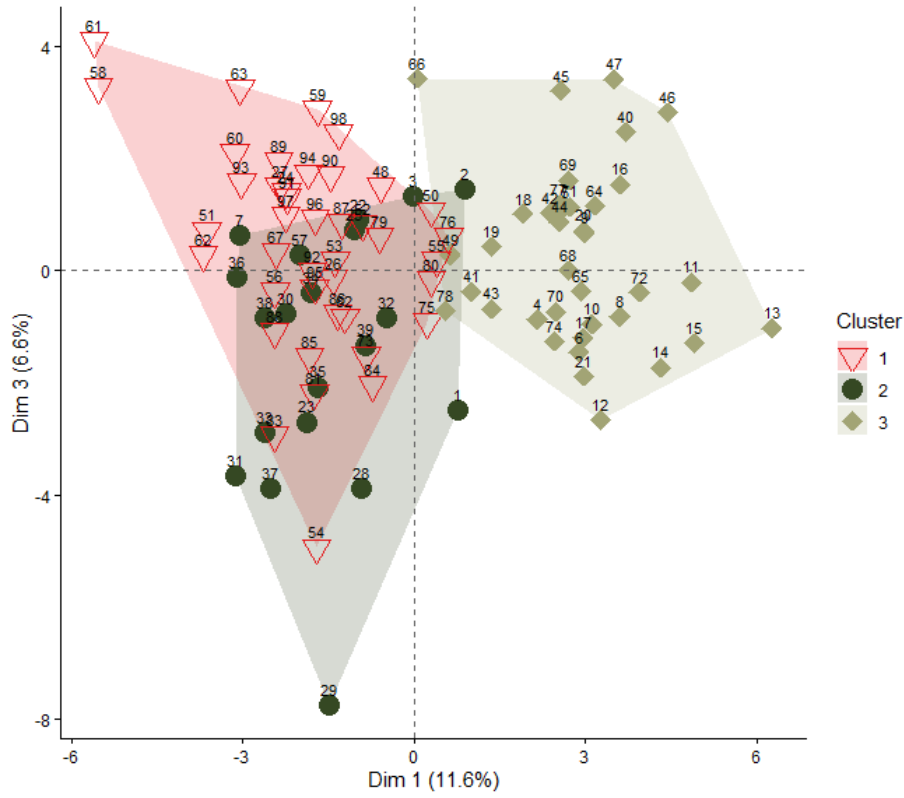


Contribution of variables to Dim-3

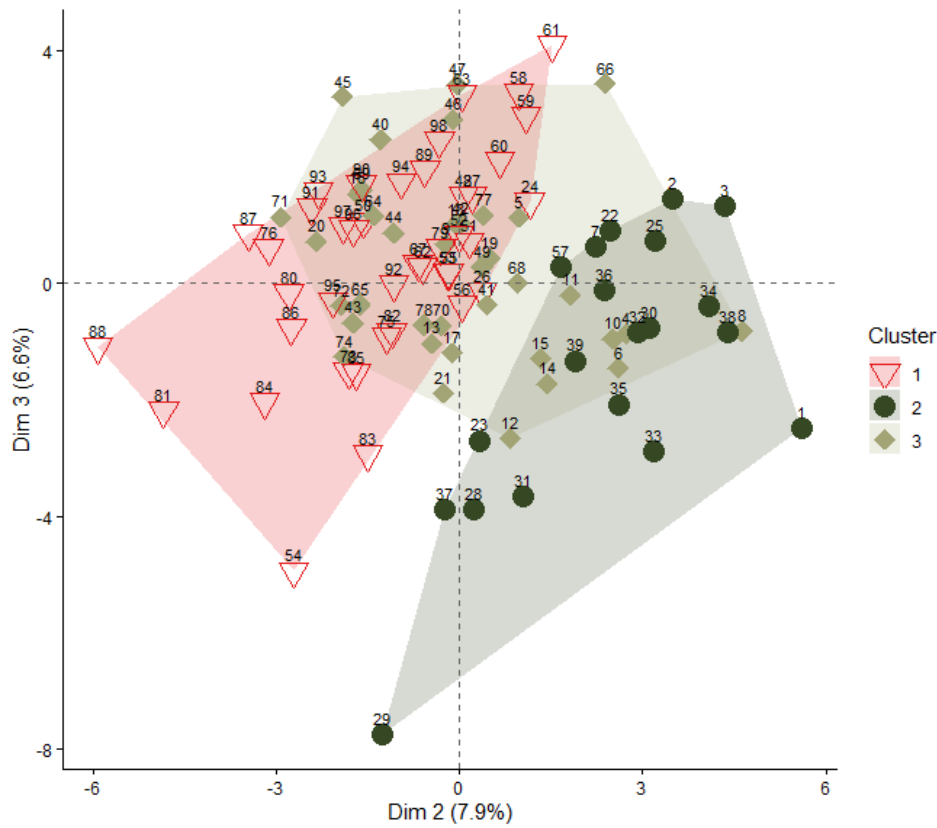


PCA plot showing the first two principal components (Dim1 and Dim2) of morphological traits for 88 plant species. The x-axis represents Dim1 (11.6%) and the y-axis represents Dim2 (7.9%). Traits are labeled as vectors originating from the center. Species are numbered 1-88. A dashed vertical line is at Dim1 = 0.

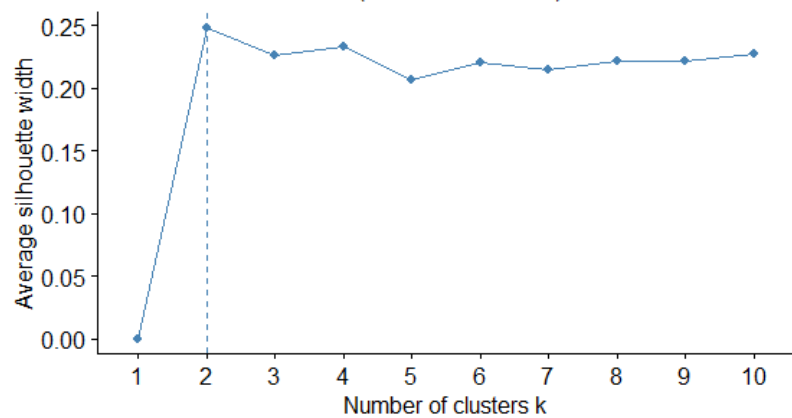
FAMD - Dim 1 & 3



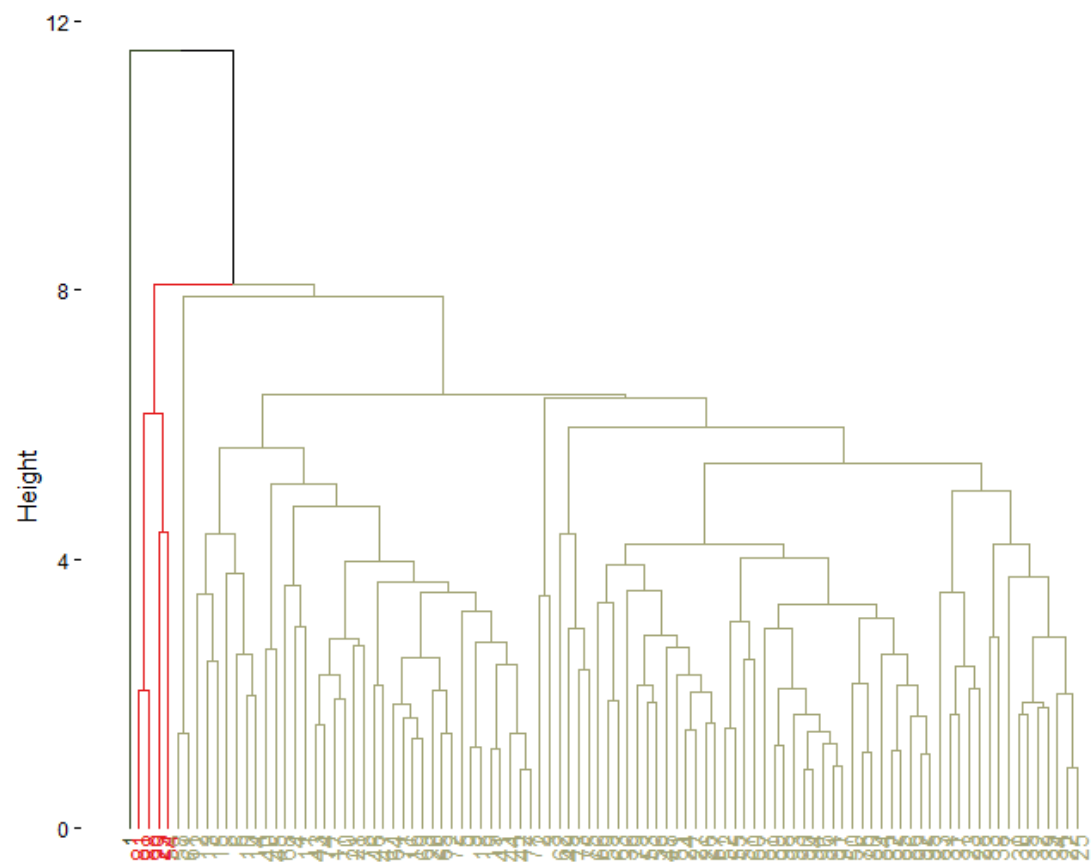
FAMD - Dim 2 & 3

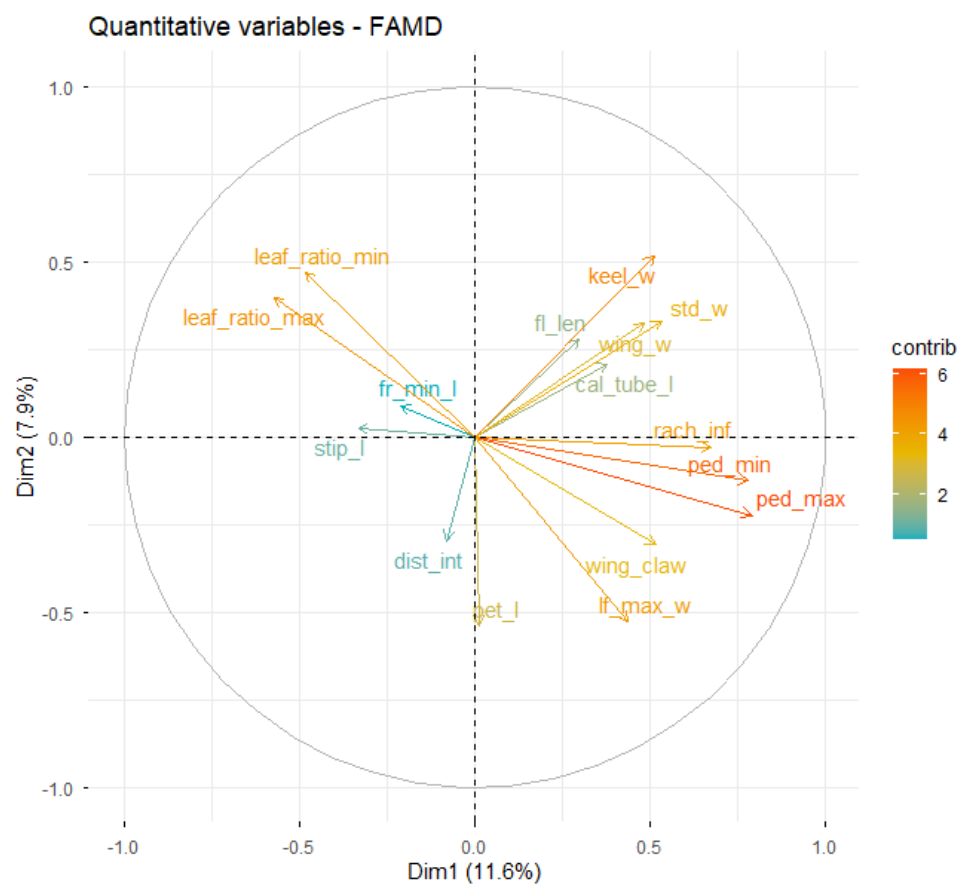
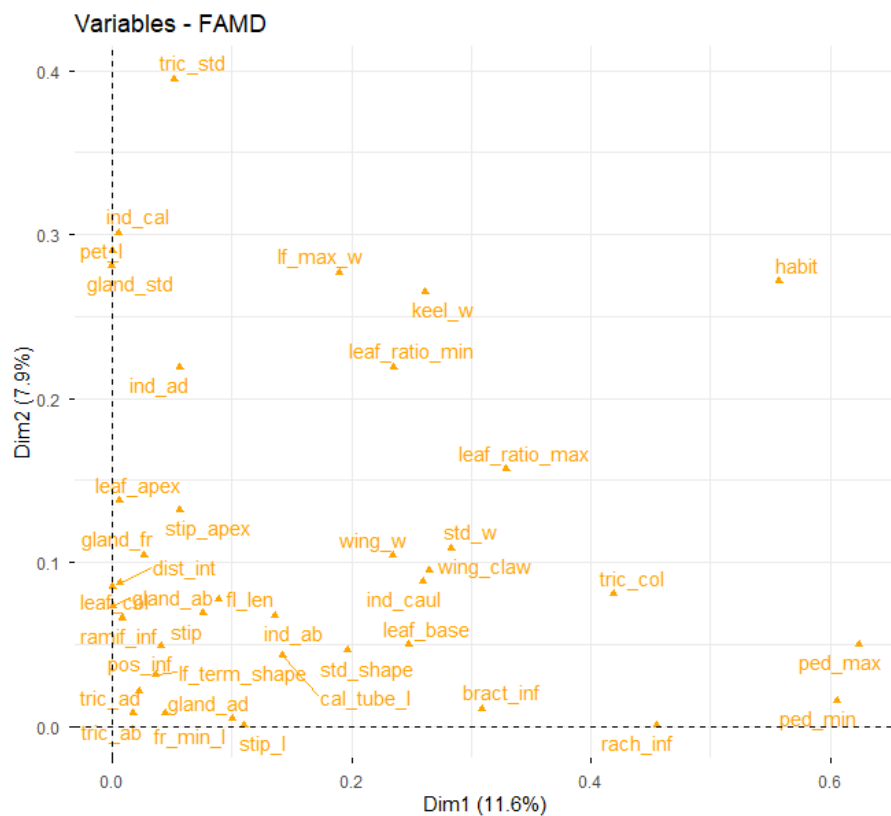


Número ótimo de clusters (método da silhueta)



UPGMA Dendrogram





Qualitative variable categories - FAMD

